

Application Portfolio & Inventory

Complete catalog of ArchiSurance application components

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1. Introduction

This document provides the definitive catalog of all application components, services, interfaces, and data objects within the ArchiSurance enterprise. It is maintained by the Application Architecture team and updated quarterly. The Enterprise Architecture Overview (EA-OVR-001) provides context on how these applications fit within the broader architecture. For infrastructure details supporting these applications, refer to the Infrastructure Design Document (EA-INFRA-003).

Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Security review was completed in accordance with the CISO's security-by-design framework. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits.

The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Integration testing spans three environments: development, staging, and pre-production. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam.

Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The architectural principles that guide this

work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Lessons learned from the pilot phase have been incorporated into the current design iteration. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024.

The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system.

A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Lessons learned from the pilot phase have been incorporated into the current design iteration. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal.

2. Application Component Catalog

ArchiSurance operates twenty application components ranging from legacy systems to modern cloud-native microservices. Each component is described below with its business alignment, technology dependencies, and data access patterns.

2.x CRM

The CRM is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 1 (business critical) in the application tiering framework.

Business processes served: Client Onboarding, Complaint Handling.

Data objects accessed: Customer Record.

Technology dependencies: Docker CE, PostgreSQL 15.

The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Lessons learned from the pilot phase have been incorporated into the current design iteration. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Security review was completed in accordance with the CISO's security-by-design framework.

Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The CRM System has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone.

The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer.

2.x PAS

The Policy Admin is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 1 (business critical) in the application tiering framework.

Business processes served: Policy Issuance, Policy Renewal.

Data objects accessed: Policy Record.

Technology dependencies: the primary relational database.

A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Integration testing spans three environments: development, staging, and pre-production. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline.

Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Stakeholders across multiple departments have contributed requirements for this workstream. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Lessons learned from the pilot phase have been incorporated into the current design iteration. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Security review was completed in accordance with the CISO's security-by-design framework. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog.

Integration testing spans three environments: development, staging, and pre-production. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel.

2.x Claims Processing System

The Claims Platform is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 1 (business critical) in the application tiering framework.

Business processes served: FNOL Process, Claim Evaluation.

Data objects accessed: Claim Record.

Technology dependencies: Kubernetes Cluster, Postgres.

Downstream application consumers: Notification Engine.

Change management activities include stakeholder communication, training sessions, and documentation updates. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside

feature development. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies.

The Claims Management Platform has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations.

ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards.

2.x Actuarial Risk Module

The Risk Scoring Engine is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 1 (business critical) in the application tiering framework.

Business processes served: Risk Evaluation, Premium Calculation.

Data objects accessed: Premium Calculation Data.

Technology dependencies: K8s.

Downstream application consumers: Guidewire PolicyCenter.

Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The design considers both current-state constraints and target-state aspirations as documented in the strategic

roadmap. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. External consultants from Deloitte provided an independent assessment of the architecture in late 2024.

The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Stakeholders across multiple departments have contributed requirements for this workstream. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana.

2.x Customer Self-Service Portal

The MyArchiSurance Portal is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Business processes served: Complaint Handling.

Technology dependencies: the in-memory cache, Nginx.

The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Change management activities include stakeholder communication, training sessions, and documentation updates. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point.

Lessons learned from the pilot phase have been incorporated into the current design iteration. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline.

2.x Document Management System

The Alfresco DMS is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Business processes served: Document Digitization, Regulatory Reporting.

Data objects accessed: Document Metadata.

Technology dependencies: Elasticsearch.

A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities.

The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The total cost of ownership analysis, prepared by the Finance Department in

collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly.

2.x Messaging Service

The Alert Service is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Technology dependencies: Kubernetes.

An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager.

Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Capacity planning exercises revealed that the

current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required.

Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period.

2.x Anti-Fraud Module

The Fraud Detection Engine is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Business processes served: Fraud Check.

Data objects accessed: Fraud Score Dataset.

Technology dependencies: the container orchestration platform.

Downstream application consumers: Claims Management Platform.

A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point.

Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The vendor selection process followed the procurement guidelines established by the CFO's office. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap.

Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch

via the CI/CD pipeline. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Lessons learned from the pilot phase have been incorporated into the current design iteration. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period.

2.x Analytics Dashboard

The Cognos Analytics is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Business processes served: Regulatory Reporting, SLA Monitoring.

Data objects accessed: Analytics Cube, Regulatory Submission Record.

Technology dependencies: Microsoft SQL Server 2022.

A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions.

The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Security review was completed in accordance with the CISO's security-by-design framework.

An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes

disruption to active users and allows incremental validation of the new implementation. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Lessons learned from the pilot phase have been incorporated into the current design iteration.

2.x MuleSoft ESB

The Enterprise Service Bus is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Technology dependencies: the event streaming platform.

Downstream application consumers: CRM System, Billing, Policy Admin.

Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point.

The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Change management activities include stakeholder communication, training sessions, and documentation updates. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days.

The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Security review was completed in accordance with the CISO's security-by-design framework. The vendor selection process followed the procurement guidelines established by the CFO's office. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days.

2.x Identity & Access Management

The IAM is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Business processes served: Customer Identity Verification.

Data objects accessed: Audit Log.

ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The vendor selection process followed the procurement guidelines established by the CFO's office. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format.

Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework.

Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. External consultants from Deloitte provided an independent assessment of the architecture in late 2024.

2.x Revenue Management System

The Revenue Management System is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Business processes served: Broker Commission Settlement, Premium Calculation.

Data objects accessed: Commission Ledger.

Technology dependencies: Oracle Database 19c.

The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Integration testing spans three environments: development, staging, and pre-production. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Lessons learned from the pilot phase have been incorporated into the current design iteration.

A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates.

The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Change management activities include stakeholder communication, training sessions, and documentation updates. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment.

2.x Broker Portal

The Broker Portal is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Business processes served: Broker Commission Settlement.

Technology dependencies: Docker.

The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The Broker Portal has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes,

include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format.

The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Security review was completed in accordance with the CISO's security-by-design framework. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Stakeholders across multiple departments have contributed requirements for this workstream. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner.

Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development.

2.x Data Warehouse

The Data Warehouse is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Data objects accessed: Analytics Cube, Policy Record, Claim Record.

Technology dependencies: Oracle RDBMS.

Downstream application consumers: Cognos Analytics.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline.

The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this

document must be reflected there within five business days. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024.

Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership.

2.x Process Automation Engine

The BPM Engine is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Business processes served: Claims Assessment, Policy Issuance.

Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Security review was completed in accordance with the CISO's security-by-design framework.

Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version

control system. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards.

2.x Gateway

The Gateway is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Technology dependencies: Kubernetes, the in-memory cache.

Downstream application consumers: MyArchiSurance Portal, Mobile App Backend, Broker Portal.

From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Change management activities include stakeholder communication, training sessions, and documentation updates.

Lessons learned from the pilot phase have been incorporated into the current design iteration. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA.

Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Integration testing spans three environments: development, staging, and pre-production. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing

durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios.

2.x Content Delivery Network

The Content Delivery Network is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Technology dependencies: Nginx.

Downstream application consumers: MyArchiSurance Portal.

Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee.

The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour.

The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Security review was completed in accordance with the CISO's security-by-design framework. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam.

2.x Email Service

The Email Service is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Downstream application consumers: Notification Service.

Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline.

The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The Email Service has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. Change management activities include stakeholder communication, training sessions, and documentation updates.

Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Lessons learned from the pilot phase have been incorporated into the current design iteration. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards.

2.x Mobile App Backend

The Mobile App Backend is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Technology dependencies: the container runtime.

During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates.

Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership.

Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users.

A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Security review was completed in accordance with the CISO's security-by-design framework.

2.x Batch Processing Engine

The Batch Processing Engine is a core component of the ArchiSurance application landscape. It was first deployed in 2022 and has undergone multiple iterations since then. The system is classified as Tier 2 (important) in the application tiering framework.

Technology dependencies: the event streaming platform.

Downstream application consumers: Oracle DWH.

Security review was completed in accordance with the CISO's security-by-design framework. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline.

The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys,

patch levels, and relationships between configuration items. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The Batch Processing Engine has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days.

A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The vendor selection process followed the procurement guidelines established by the CFO's office. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Integration testing spans three environments: development, staging, and pre-production. The Batch Processing Engine has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture.

3. Application Services

Application services represent the externally visible behavior of application components. Each service is exposed through one or more interfaces and realized by specific components.

Policy API

The Policy API is realized by: Guidewire PolicyCenter. This service is exposed via the REST API Gateway and authenticated through the IAM platform.

Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline.

The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards.

Claims API

The Claims API is realized by: Claims Management Platform. This service is exposed via the REST API Gateway and authenticated through the Access Management System platform.

The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period.

Security review was completed in accordance with the CISO's security-by-design framework. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Lessons learned from the pilot phase have been incorporated into the current design iteration.

Risk Scoring API

The Risk Scoring API is realized by: Risk Calculator. This service is exposed via the REST API Gateway and authenticated through the Identity & Access Management platform.

Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions.

Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures.

Customer API

The Customer API is realized by: CRM System. This service is exposed via the REST API Gateway and authenticated through the IAM platform.

Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly.

Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog.

Notification API

The Notification API is realized by: Messaging Service. This service is exposed via the REST API Gateway and authenticated through the Access Management System platform.

From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Change management activities include stakeholder communication, training sessions, and documentation updates. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval.

The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures.

Document API

The Document API is realized by: Alfresco DMS. This service is exposed via the REST API Gateway and authenticated through the Okta IAM platform.

Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits.

The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Lessons learned from the pilot phase have been incorporated into the current design iteration. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap.

Fraud Scoring API

The Fraud Scoring API is realized by: FDE. This service is exposed via the REST API Gateway and authenticated through the IAM platform.

The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana.

Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap.

Payment API

The Payment API is realized by: Billing Engine. This service is exposed via the REST API Gateway and authenticated through the Okta IAM platform.

The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment.

Reporting API

The Reporting API is realized by: Reporting & Analytics Platform. This service is exposed via the REST API Gateway and authenticated through the Access Management System platform.

External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly.

Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch.

via the CI/CD pipeline.

Authentication API

The Authentication API is realized by: Okta IAM. This service is exposed via the REST API Gateway and authenticated through the IAM platform.

The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Lessons learned from the pilot phase have been incorporated into the current design iteration. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024.

4. Application Interfaces

REST API Gateway

The REST API Gateway provides external connectivity for partners, customers, and internal consumers. As described in the Integration Architecture document (EA-INT-007), all interfaces are secured with TLS 1.3 and require OAuth 2.0 bearer tokens.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model.

Mobile App Interface

The Mobile App Interface provides external connectivity for partners, customers, and internal consumers. As described in the Integration Architecture document (EA-INT-007), all interfaces are secured with TLS 1.3 and require OAuth 2.0 bearer tokens.

A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging

environment to validate the resilience of the system against network partitions and dependency failures. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The vendor selection process followed the procurement guidelines established by the CFO's office. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana.

Partner Integration Hub

The Partner Integration Hub provides external connectivity for partners, customers, and internal consumers. As described in the Integration Architecture document (EA-INT-007), all interfaces are secured with TLS 1.3 and require OAuth 2.0 bearer tokens.

The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format.

SOAP Legacy Interface

The SOAP Legacy Interface provides external connectivity for partners, customers, and internal consumers. As described in the Integration Architecture document (EA-INT-007), all interfaces are secured with TLS 1.3 and require OAuth 2.0 bearer tokens.

The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025.

GraphQL Analytics Endpoint

The GraphQL Analytics Endpoint provides external connectivity for partners, customers, and internal consumers. As described in the Integration Architecture document (EA-INT-007), all interfaces are secured with TLS 1.3 and require OAuth 2.0 bearer tokens.

The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates.

5. Data Objects

Policy Record

The Policy Record is a critical data asset. It is accessed by: PAS, Oracle DWH. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager.

Claim Record

The Claim Record is a critical data asset. It is accessed by: CMP, Oracle DWH. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval.

Customer Record

The Customer Record is a critical data asset. It is accessed by: Salesforce CRM. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits.

Premium Calculation Data

The Premium Calculation Data is a critical data asset. It is accessed by: Risk Calculator. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Capacity planning

exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required.

Fraud Score Dataset

The Fraud Score Dataset is a critical data asset. It is accessed by: Fraud Engine. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer.

Commission Ledger

The Commission Ledger is a critical data asset. It is accessed by: Billing System. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Integration testing spans three environments: development, staging, and pre-production. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap.

Audit Log

The Audit Log is a critical data asset. It is accessed by: Access Management System. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Lessons learned from the pilot phase have been incorporated into the current design iteration.

Regulatory Submission Record

The Regulatory Submission Record is a critical data asset. It is accessed by: BI Platform. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline.

Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points.

Document Metadata

The Document Metadata is a critical data asset. It is accessed by: Document Repository. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership.

Analytics Cube

The Analytics Cube is a critical data asset. It is accessed by: Analytics Dashboard, DWH. Data governance policies for this object are described in the Data Architecture and Governance document (EA-DAT-008).

Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation.

6. Integration Map

The integration map shows data flows between application components. The ESB serves as the primary integration backbone, complemented by the API Gateway for RESTful interactions.

The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The vendor selection process followed the procurement guidelines established by the CFO's office. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The application integration has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Security review was completed in accordance with the CISO's security-by-design framework. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Integration testing spans three environments: development, staging, and pre-production. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation.

The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Change management activities include stakeholder communication, training sessions, and documentation updates. Security review was completed in accordance with the CISO's security-by-design framework. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation.

A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The application integration patterns has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Change management activities include stakeholder communication, training sessions, and documentation updates. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval.

The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Change management activities include stakeholder communication, training sessions, and documentation updates. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee.

7. Ownership and Support Matrix

Application	Owner	Tier	Support Team
CRM System	Claims IT	T1	Team-A
Policy Administration Sys	Policy IT	T1	Team-B
Claims Management Platfor	Platform Eng	T1	Team-C
Risk Engine	Data Team	T1	Team-D
Customer Self-Service Por	Infra Ops	T1	Team-E
Document Management Syste	Security	T2	Team-F
Notification Service	Claims IT	T2	Team-A
Fraud Detection Engine	Policy IT	T2	Team-B
Reporting & Analytics Pla	Platform Eng	T2	Team-C
Enterprise Service Bus	Data Team	T2	Team-D
Identity & Access Managem	Infra Ops	T2	Team-E
Billing System	Security	T2	Team-F
Broker Portal	Claims IT	T2	Team-A
Data Warehouse	Policy IT	T2	Team-B
Workflow Engine	Platform Eng	T2	Team-C
API Gateway	Data Team	T2	Team-D
Content Delivery Network	Infra Ops	T2	Team-E
Email Service	Security	T2	Team-F
Mobile App Backend	Claims IT	T2	Team-A
Batch Processing Engine	Policy IT	T2	Team-B

The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The vendor selection process followed the procurement guidelines established by the CFO's office. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel.

The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The vendor selection process followed the procurement guidelines established by the CFO's office. Change management activities include stakeholder communication, training sessions, and documentation updates. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which

additional compute and storage resources will be required. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates.

From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits.

8. Appendices

References and Related Documents

Doc ID	Title	Version
EA-OVR-001	Enterprise Architecture Overview	3.0
EA-APP-002	Application Portfolio & Inventory	4.2
EA-INFRA-003	Infrastructure Design Document	2.5
EA-BIZ-004	CRM Business Process Description	2.0
EA-CLM-005	Claims Processing Workflow	3.1
EA-SEC-006	Security Architecture	2.3
EA-INT-007	Integration Architecture	2.8
EA-DAT-008	Data Architecture & Governance	2.0
EA-CLD-009	Cloud Migration Assessment	1.5
EA-DR-010	Disaster Recovery & Business Contin	3.0
EA-NET-011	Network Architecture	2.2
EA-ROAD-012	Strategic IT Roadmap 2025-2027	2.0
EA-SVC-013	IT Service Catalog	3.5
EA-VND-014	Vendor & Technology Assessment	1.8
EA-OPS-015	Operational Runbook: Core Platform	4.1
EA-CMP-016	Compliance & Regulatory Requirement	2.5
EA-CHG-017	Change Management: ERP Migration	1.2

Distribution List

Name	Role	Responsibility
J. van den Berg	Enterprise Architect	Review & Approve
M. de Vries	Application Architect	Review & Contribute
P. Jansen	Integration Architect	Review & Contribute
K. Bakker	Infrastructure Architect	Review & Contribute
S. Visser	Business Analyst	Inform
T. Mulder	Process Architect	Inform
R. de Jong	CISO	Review & Approve
L. Hendriks	Data Architect	Inform
A. Willems	CTO	Final Approval
B. Smit	Head of Claims	Business Sign-off

Glossary of Acronyms

Acronym	Definition
API	Application Programming Interface
BPM	Business Process Management
CDN	Content Delivery Network
CISO	Chief Information Security Officer
CRM	Customer Relationship Management
DR	Disaster Recovery
DMS	Document Management System
DMZ	Demilitarized Zone
EA	Enterprise Architecture
ESB	Enterprise Service Bus
ETL	Extract, Transform, Load

FNOL	First Notice of Loss
GDPR	General Data Protection Regulation
HSM	Hardware Security Module
IAM	Identity and Access Management
KPI	Key Performance Indicator
LAN	Local Area Network
MTTR	Mean Time to Recovery
NPS	Net Promoter Score
PAS	Policy Administration System
PCI-DSS	Payment Card Industry Data Security Standard
PoC	Proof of Concept
RBAC	Role-Based Access Control
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SAN	Storage Area Network
SLA	Service Level Agreement
SOX	Sarbanes-Oxley Act
STP	Straight-Through Processing
VLAN	Virtual Local Area Network
VPN	Virtual Private Network
WAN	Wide Area Network

Detailed Revision Notes for Version 2.1

Section 2.3 was updated to reflect the new naming convention adopted by the Platform Engineering team. All references to 'microservice cluster' have been replaced with 'application workload group' per the Architecture Board decision of January 2025. Section 4.1 was expanded to include the results of the capacity planning exercise conducted in Q4 2024, which identified the need for additional compute resources in the Amsterdam primary data center.

Appendix B was added to include the RACI matrix requested by the PMO during the Q3 2024 quarterly review. The glossary was updated to include twelve new acronyms that were introduced since the previous version. Figure 3 was replaced with an updated diagram generated from the Sparx EA repository on March 28, 2025. The document classification was elevated from 'Internal' to 'Internal - Confidential' following the data classification review of February 2025.

Feedback from the peer review conducted by M. de Vries on March 10, 2025, addressed three items: (1) clarification of the boundary between the DMZ and the application tier, (2) addition of the backup retention policy reference, and (3) correction of the SLA target for the Claims Handling Service from 99.5% to 99.9% as agreed in the contract amendment of December 2024.

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